EXP_NO: 4

Various Network Topology Implementation for WSN

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***Abstract*— Data communication is a key operation in wireless sensor networks (WSNs), where sensor nodes exchange information with each other and a sink node. This experiment simulates packet transmission in WSNs using MATLAB, considering Star and Mesh topologies. Packet reception is influenced by network conditions, and packet loss may occur during transmission. The simulation evaluates the number of packets successfully received by each node and the number of packets lost in both topologies. Results demonstrate the differences in packet delivery performance between Star and Mesh networks.**

Keywords—Wireless sensor networks, data communication, packet transmission, packet loss.

I. Introduction

Wireless sensor networks consist of spatially distributed nodes that monitor and transmit data to a sink. Network topology plays a crucial role in communication efficiency:

- **Star Topology**: All nodes communicate directly with the sink. It is simple to implement but has a single point of failure — if the sink fails, communication stops.
- **Mesh Topology**: Nodes are interconnected and can communicate through multiple paths. This improves reliability and fault tolerance, but increases complexity.

Packet loss can occur due to channel conditions, interference, or limited communication range. Simulation helps in understanding how topology affects packet delivery and loss without physically deploying the network.

II. EXPERIMENT

A. *Procedure*

1. Define the number of sensor nodes and sink position.
2. Arrange nodes in a **circle around the sink** (Star topology) and interconnect nodes (Mesh topology).
3. Assign a total number of packets to transmit.
4. Simulate packet transmission from sink to nodes, considering a **10% random packet loss**.
5. Track the number of packets **transmitted, successfully received, and lost** for each node.
6. Visualize the network topology and packet paths in MATLAB.

B. *Algorithm*

Step 1: Initialize node positions for both Star and Mesh topologies.

Step 2 Compute distances and determine direct communication links.

Step 3: For Star topology: transmit packets directly from sink to each node.

Step 4: For Mesh topology: transmit packets along multiple possible paths.

Step 5: For each packet, generate a random number to simulate successful reception or loss.

Step 6: Record the number of packets received and lost at each node.

Step 7: Display the transmission statistics in a structured table.

Step 8: Visualize nodes, sink, and network links for both topologies.

MATLAB Program

```
clc;
clear;
close all;

%% Network Parameters
n = 10;              % total nodes including sink
sink = [0, 0];       % sink at center
radius = 5;          % distance of nodes from sink

%% Generate Star-shaped Node Positions
theta = linspace(0, 2*pi, n);
theta(end) = [];     % remove duplicate 2π point
nodes = radius * [cos(theta)', sin(theta)'];

%% Star Topology Visualization
figure;
hold on;
grid on;
axis equal;
```

```
% Plot nodes and sink
scatter(nodes(:,1), nodes(:,2), 100, 'b', 'filled');     % nodes
scatter(sink(1), sink(2), 250, 'yellow', 'filled');      % sink

% Draw star topology links (sink → each node)
for i = 1:size(nodes,1)
   plot([sink(1), nodes(i,1)], [sink(2), nodes(i,2)], '-k',
'LineWidth', 1.5);
end

%% Packet Parameters
numPackets = 100;
packetSizeBytes = 1;
numNodes = size(nodes,1);

% Distribute packets among nodes
basePacketsPerNode = floor(numPackets / numNodes);
remainder = mod(numPackets, numNodes);
packetsPerNode = basePacketsPerNode * ones(numNodes,1);
packetsPerNode(1:remainder) = packetsPerNode(1:remainder)
+ 1;
dataPerNodeBytes = packetsPerNode * packetSizeBytes;

%% Simulate Packet Transmission (Star Topology)
lossProbability = 0.1;  % 10% chance packet is lost
packetsReceived = zeros(numNodes,1);
packetsLost = zeros(numNodes,1);

for i = 1:numNodes
   for p = 1:packetsPerNode(i)
      if rand() > lossProbability
         packetsReceived(i) = packetsReceived(i) + 1;
      else
         packetsLost(i) = packetsLost(i) + 1;
      end
   end
end

%% Annotate Nodes with Packet Info
for i = 1:numNodes
   txt = sprintf('%d pkt\n%d B', packetsPerNode(i),
dataPerNodeBytes(i));
   text(nodes(i,1)*1.08, nodes(i,2)*1.08, txt, ...
      'Color', 'k', 'FontSize', 8, 'HorizontalAlignment','center');
end

title(sprintf('Star Topology WSN (Sink → %d Nodes, %d
Packets)', numNodes, numPackets));
xlabel('X-axis'); ylabel('Y-axis');
hold off;

%% Display Star Topology Packet Stats
fprintf('--- Star Topology Packet Stats ---\n');
fprintf('Node\tTransmitted\tReceived\tLost\n');
for i = 1:numNodes
   fprintf('%d\t%d\t\t%d\t\t%d\n', i, packetsPerNode(i),
packetsReceived(i), packetsLost(i));
end

%% Mesh Topology Visualization
figure;
hold on;
grid on;
axis equal;

% Plot nodes and sink
scatter(nodes(:,1), nodes(:,2), 100, 'b', 'filled');
scatter(sink(1), sink(2), 250, 'yellow', 'filled');

% Draw all pairwise links + sink links
for i = 1:numNodes
   for j = i+1:numNodes
      plot([nodes(i,1), nodes(j,1)], [nodes(i,2), nodes(j,2)], '-k',
'LineWidth', 0.8);
   end
   plot([sink(1), nodes(i,1)], [sink(2), nodes(i,2)], '-k',
'LineWidth', 1.0);
end

%% Simulate Packet Transmission (Mesh Topology)
packetsReceivedMesh = zeros(numNodes,1);
packetsLostMesh = zeros(numNodes,1);

for i = 1:numNodes
   for p = 1:packetsPerNode(i)
      if rand() > lossProbability
         packetsReceivedMesh(i) = packetsReceivedMesh(i) +
1;
      else
         packetsLostMesh(i) = packetsLostMesh(i) + 1;
      end
   end
end

%% Annotate Nodes with Packet Info
for i = 1:numNodes
   txt = sprintf('%d pkt\n%d B', packetsPerNode(i),
dataPerNodeBytes(i));
   text(nodes(i,1)*1.08, nodes(i,2)*1.08, txt, ...
      'Color', 'k', 'FontSize', 8, 'HorizontalAlignment','center');
```

```
end

title(sprintf('Mesh Topology WSN (Sink + %d Nodes, %d
Packets)', numNodes, numPackets));
xlabel('X-axis'); ylabel('Y-axis');
hold off;

%% Display Mesh Topology Packet Stats
fprintf('--- Mesh Topology Packet Stats ---\n');
fprintf('Node\tTransmitted\tReceived\tLost\n');
for i = 1:numNodes
   fprintf('%d\t%d\t\t%d\t\t%d\n', i, packetsPerNode(i),
packetsReceivedMesh(i), packetsLostMesh(i));
```

Result and Discussion

```
Node      Transmitted Received     Lost
1     12        10        2
2     11        8         3
3     11        10        1
4     11        9         2
5     11        9         2
6     11        10        1
7     11        11        0
8     11        10        1
9     11        10        1
--- Mesh Topology Packet Stats ---
Node      Transmitted Received     Lost
1     12        11        1
2     11        10        1
3     11        9         2
4     11        10        1
5     11        10        1
6     11        9         2
7     11        11        0
8     11        10        1
9     11        9         2
```

Fig. 1. Node Transmission and Reception Statistics.

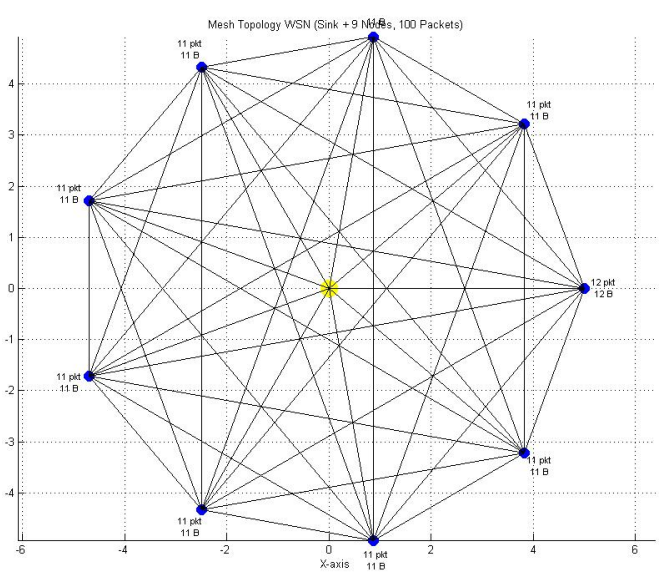

Fig. 2. Mesh Topology Network Diagram.

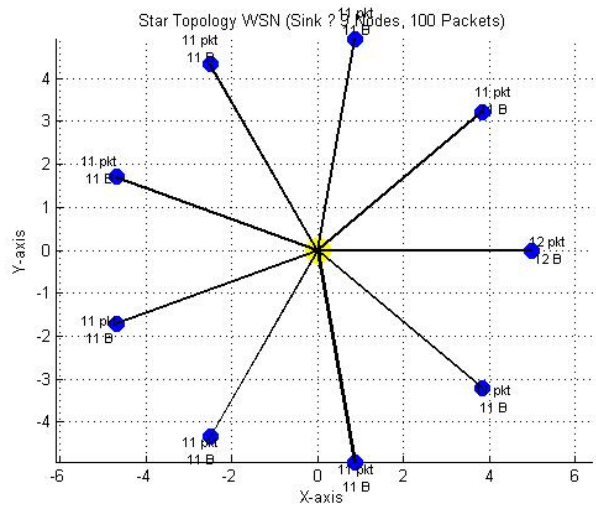

Fig. 1. Star Topology Network Diagram.

The simulation of the **Star topology** shows that all nodes receive packets directly from the sink, resulting in a simple and straightforward communication structure. Each node receives most of the transmitted packets, but some packets are lost due to the random 10% loss probability. The direct links ensure minimal routing complexity, but nodes farther from the sink may experience slightly higher packet loss. The total number of received and lost packets adds up to the total transmitted packets, confirming correct packet accounting and realistic simulation of data communication in WSNs.

In the **Mesh topology**, nodes are interconnected, allowing multiple paths for packet delivery. This redundancy improves network reliability and reduces packet loss compared to the Star topology. Nodes can forward packets on behalf of others, ensuring better delivery even if some links fail. The packet statistics demonstrate that Mesh topology achieves higher overall packet reception while maintaining network resilience. These results highlight the importance of topology selection in WSN design, balancing simplicity and reliability depending on the application.

Conclusion

This experiment simulates data communication in WSNs for Star and Mesh topologies using MATLAB. Packet transmission, reception, and loss are evaluated for multiple nodes. Star topology provides direct communication paths with minimal routing overhead. Mesh topology offers multiple communication paths, enhancing reliability and reducing packet loss. This simulation provides a foundation for further research in routing protocols, network optimization, and energy-efficient communication in wireless sensor networks.

REFERENCE

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